Skill - 3

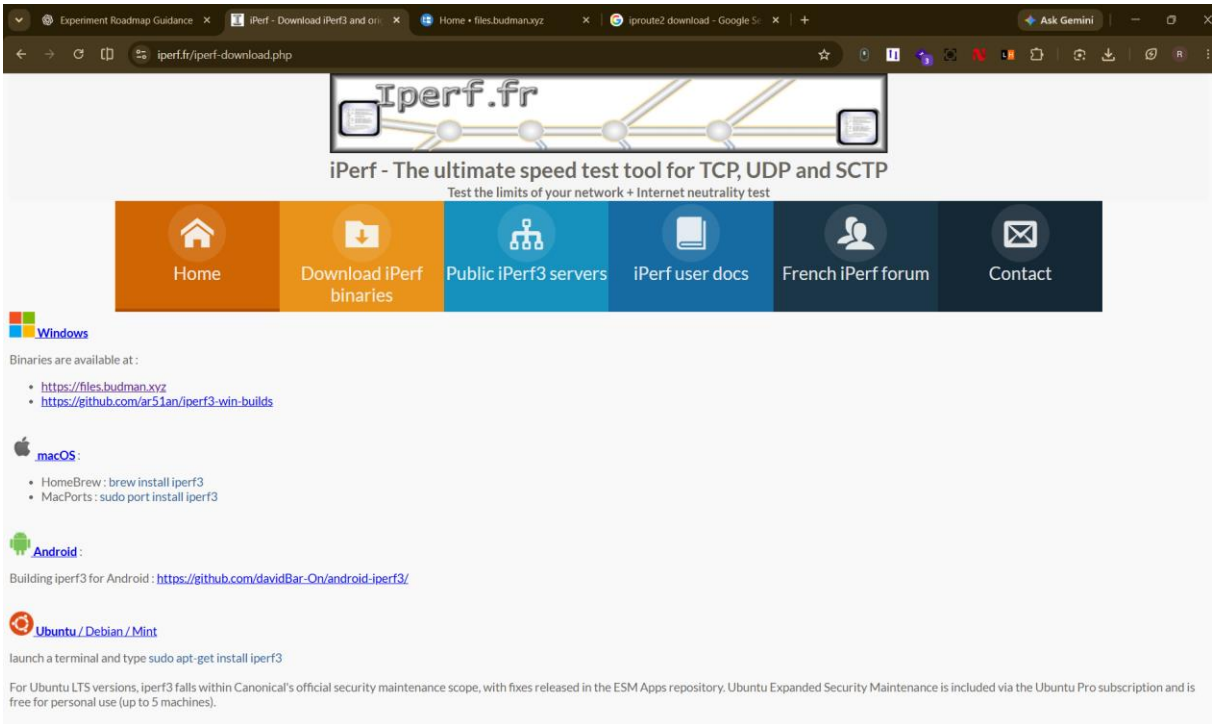
Raghuram Reddy

Section – 11

2520090024

To study network queueing behavior, identify the queueing threshold (knee), examine buffer behavior under different traffic loads, and experimentally validate Little's Law using packet measurements.

1. **Install the software**




```
raghuram32@raghuramreddy:~$ sudo apt update
sudo apt install iperf3 -y
[sudo: authenticate] Password:
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Err:2 http://security.ubuntu.com/ubuntu resolute-security Release
  Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4000:1::102). - connect (101: Network is unreachable
) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::102). - connect (101: Network is unreachable
) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:1000::18). - connect (101: Network is unreachabl
e) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:10:1000::216). - connect (101: Network is unreach
able) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::101). - connect (101: Network is unreach
able) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:10:1000::35c). - connect (101: Network is unre
achable) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:10:1000::398). - connect (101: Network is u
nreachable) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:1000::16). - connect (101: Network is
unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:1000::17). - connect (101: Network is
 unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4000:1::103). - connect (101: Network is
 unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::103). - connect (101: Network is
 unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4000:1::101). - connect (101: Network is
 unreachable)
Hit:3 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [868 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [389 kB]
Error: The repository 'http://security.ubuntu.com/ubuntu resolute-security Release' no longer has a Release file.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
Notice: See apt-secure(8) manpage for repository creation and user configuration details.
iperf3 is already the newest version (3.20-2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 76
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo apt -o Acquire::ForceIPv4=true update
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [717 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [220 kB]
Fetched 1075 kB in 3s (324 kB/s)
76 packages can be upgraded. Run 'apt list --upgradable' to see them.
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ iperf3 --version
iperf 3.20 (cJSON 1.7.15)
Linux raghuramreddy 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64
Optional features available: CPU affinity setting, IPv6 flow label, SCTP, TCP congestion algorithm setting, sendfile / z
erocopy, socket pacing, authentication, bind to device, support IPv4 don't fragment, POSIX threads
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ iperf3 --version
iperf 3.20 (cJSON 1.7.15)
Linux raghuramreddy 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64
Optional features available: CPU affinity setting, IPv6 flow label, SCTP, TCP congestion algorithm setting, sendfile / z
erocopy, socket pacing, authentication, bind to device, support IPv4 don't fragment, POSIX threads
raghuram32@raghuramreddy:~$ ping -4 -c 4 google.com
PING google.com (142.251.222.206) 56(84) bytes of data.
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=1 ttl=112 time=43.3 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=2 ttl=112 time=64.1 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=3 ttl=112 time=52.6 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=4 ttl=112 time=57.4 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 43.307/54.338/64.088/7.565 ms
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo apt update
sudo apt install iperf3 -y
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
76 packages can be upgraded. Run 'apt list --upgradable' to see them.
iperf3 is already the newest version (3.20-2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 76
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ iperf3 --version
iperf 3.20 (cJSON 1.7.15)
Linux raghuramreddy 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64
Optional features available: CPU affinity setting, IPv6 flow label, SCTP, TCP congestion algorithm setting, sendfile / z
erocopy, socket pacing, authentication, bind to device, support IPv4 don't fragment, POSIX threads
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ sudo apt update
sudo apt install iproute2 -y
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
76 packages can be upgraded. Run 'apt list --upgradable' to see them.
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 76
raghuram32@raghuramreddy:~$
```

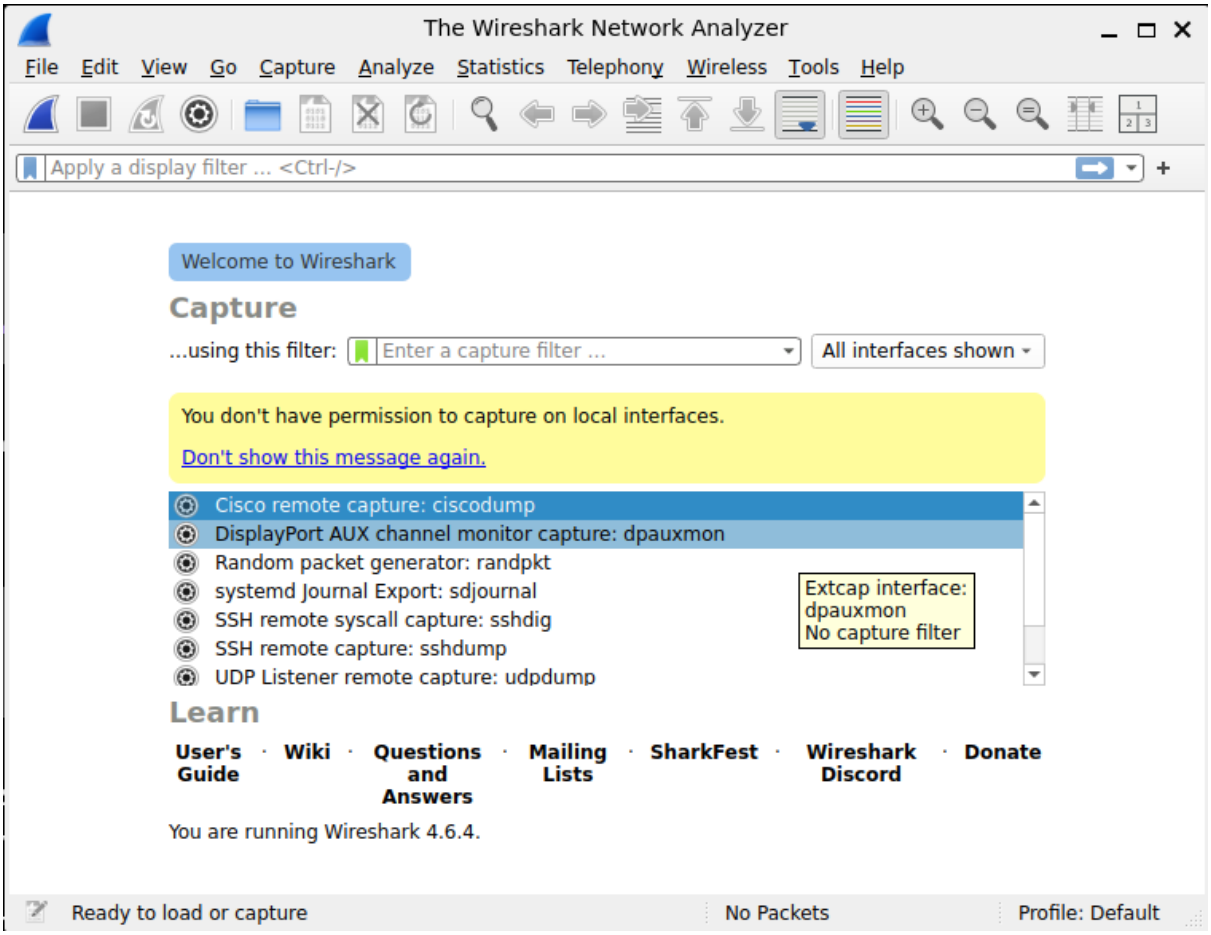
```
raghuram32@raghuramreddy:~$ iperf3 --version
iperf 3.20 (cJSON 1.7.15)
Linux raghuramreddy 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64
Optional features available: CPU affinity setting, IPv6 flow label, SCTP, TCP congestion algorithm setting, sendfi
erocopy, socket pacing, authentication, bind to device, support IPv4 don't fragment, POSIX threads
raghuram32@raghuramreddy:~$ sudo apt update
sudo apt install iproute2 -y
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
76 packages can be upgraded. Run 'apt list --upgradable' to see them.
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 76
raghuram32@raghuramreddy:~$ tc -V
tc utility, iproute2-6.19.0, libbpf 1.6.3
raghuram32@raghuramreddy:~$
```

```
raghuram32@raghuramreddy:~$ tc -V
tc utility, iproute2-6.19.0, libbpf 1.6.3
raghuram32@raghuramreddy:~$ sudo apt update
sudo apt install wireshark -y
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
76 packages can be upgraded. Run 'apt list --upgradable' to see them.
wireshark is already the newest version (4.6.4-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 76
raghuram32@raghuramreddy:~$
```

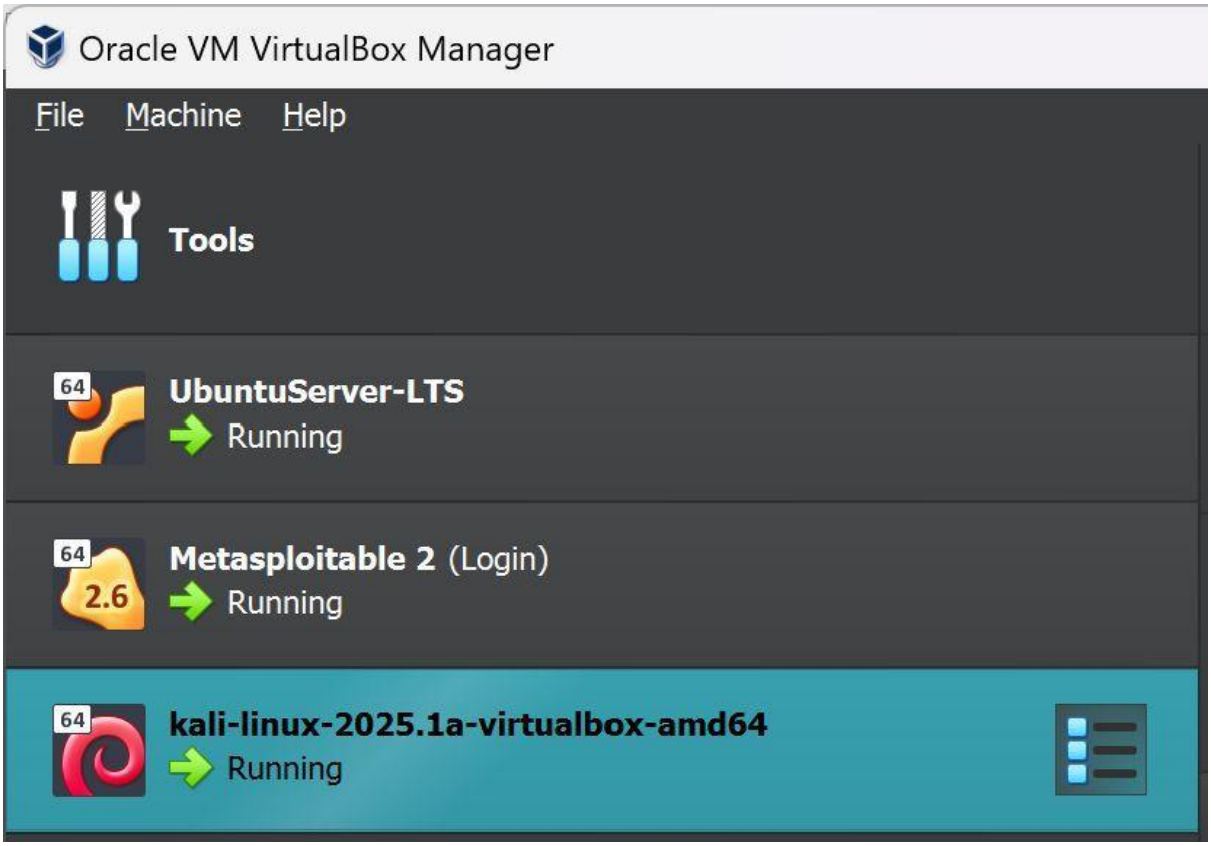
```
raghuram32@raghuramreddy:~$ sudo usermod -aG wireshark $raghuram32
Usage: usermod [options] LOGIN

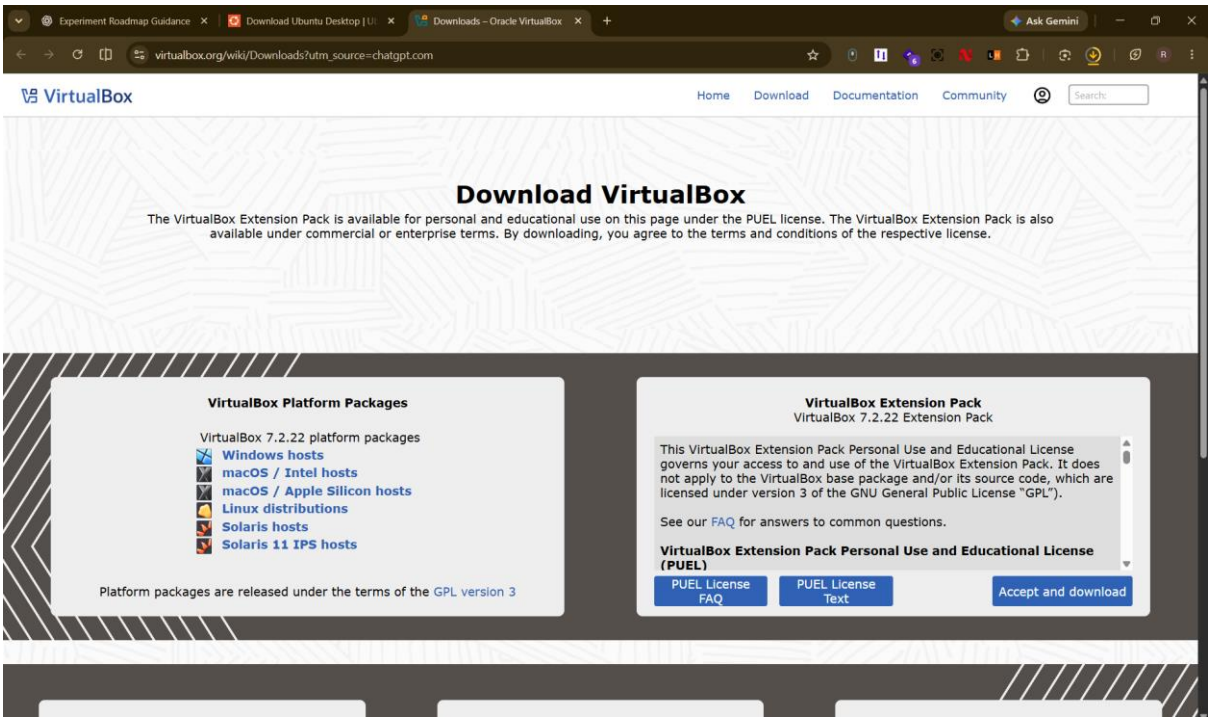
Options:
  -a, --append                  append the user to the supplemental GROUPS
                                mentioned by the -G option without removing
                                the user from other groups
  -b, --badname                 allow bad names (DEPRECATED)
  -c, --comment COMMENT         new value of the GECOS field
  -d, --home HOME_DIR           new home directory for the user account
  -e, --expiredate EXPIRE_DATE  set account expiration date to EXPIRE_DATE
  -f, --inactive INACTIVE       set password inactive after expiration
                                to INACTIVE
  -g, --gid GROUP               force use GROUP as new primary group
  -G, --groups GROUPS           new list of supplementary GROUPS
  -h, --help                    display this help message and exit
  -l, --login NEW_LOGIN         new value of the login name
  -L, --lock                    lock the user account
  -m, --move-home               move contents of the home directory to the
                                new location (use only with -d)
  -o, --non-unique              allow using duplicate (non-unique) UID
  -p, --password PASSWORD       use encrypted password for the new password
  -P, --prefix PREFIX_DIR       prefix directory where are located the /etc/* files
  -r, --remove                  remove the user from only the supplemental GROUPS
                                mentioned by the -G option without removing
                                the user from other groups
  -R, --root CHROOT_DIR         directory to chroot into
  -s, --shell SHELL             new login shell for the user account
  -u, --uid UID                 new UID for the user account
  -U, --unlock                  unlock the user account
  -v, --add-subuids FIRST-LAST  add range of subordinate uids
  -V, --del-subuids FIRST-LAST  remove range of subordinate uids
  -w, --add-subgids FIRST-LAST  add range of subordinate gids
  -W, --del-subgids FIRST-LAST  remove range of subordinate gids
  -Z, --selinux-user SEUSER     new SELinux user mapping for the user account
      --selinux-range SERANGE   new SELinux MLS range for the user account

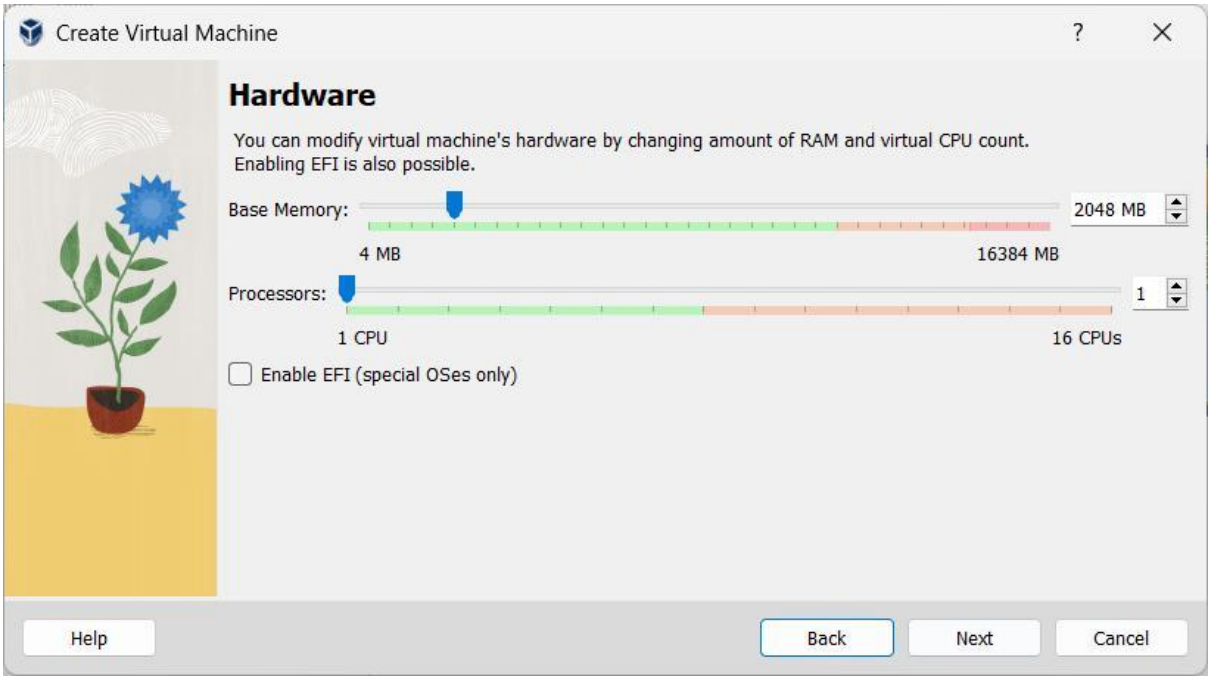
raghuram32@raghuramreddy:~$
```

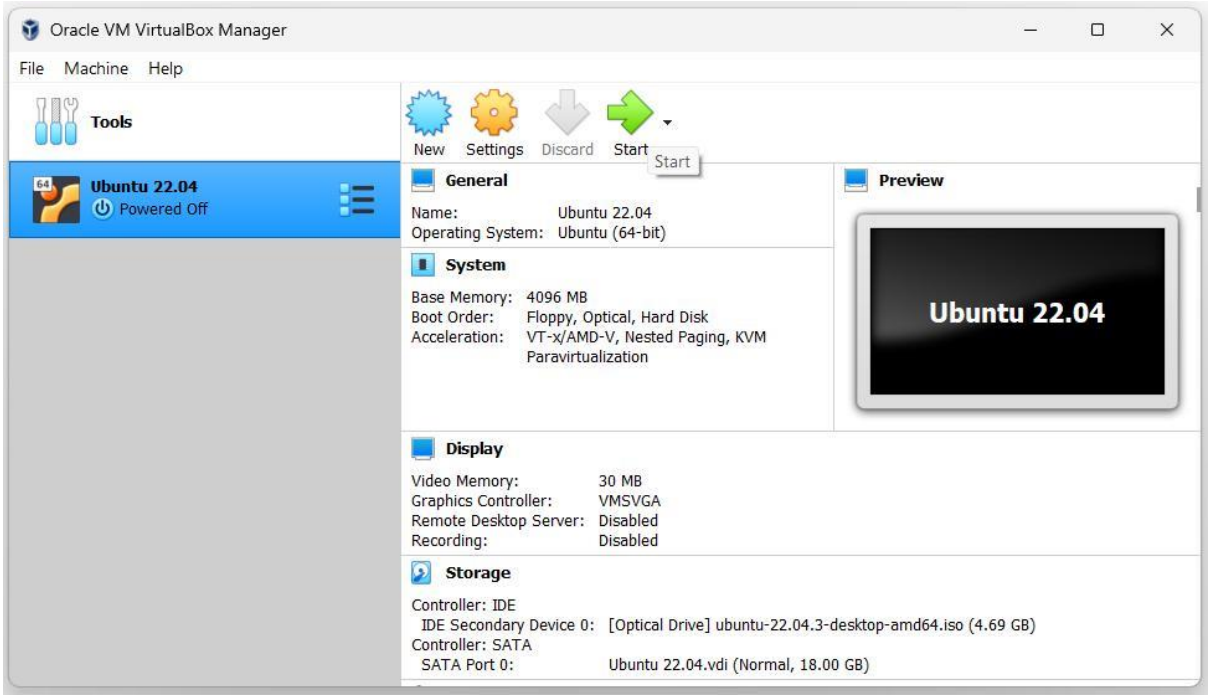


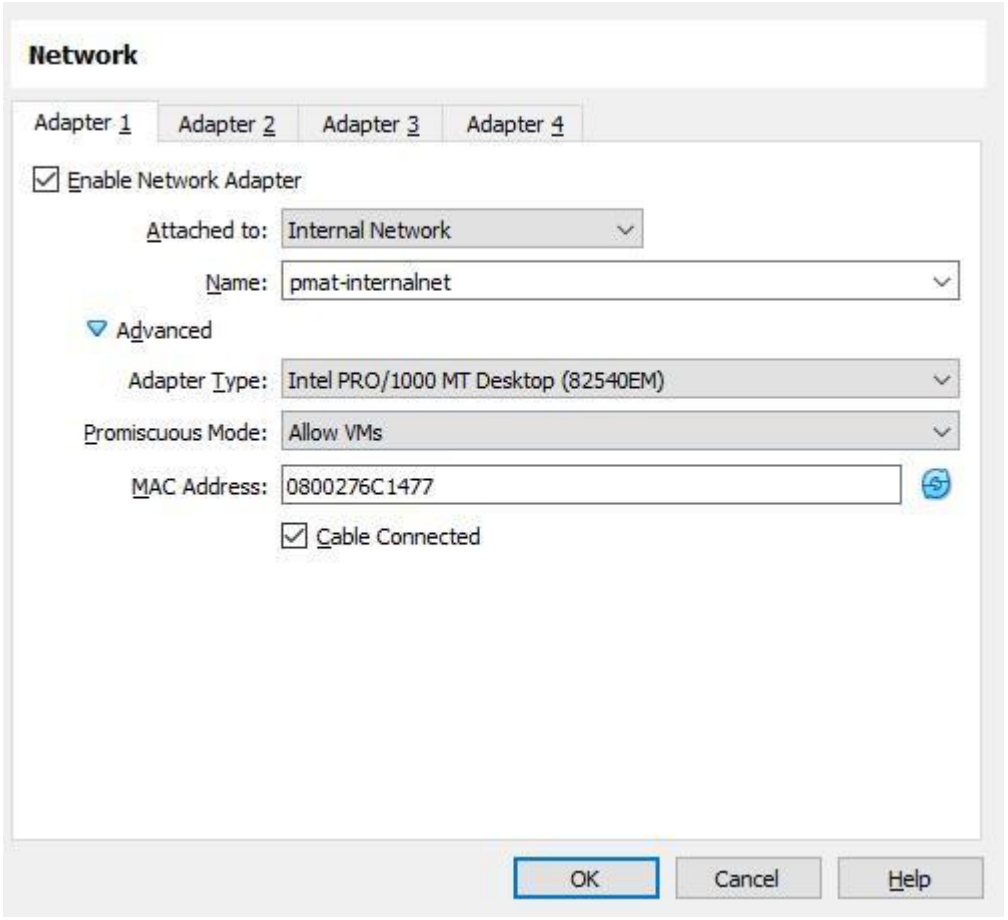

```
raghuram32@raghuramreddy:~$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 10.255.255.254/32 ::1/128
eth0             UP             172.18.167.71/20 fe80::215:5dff:fe3c:527b/64
raghuram32@raghuramreddy:~$
```

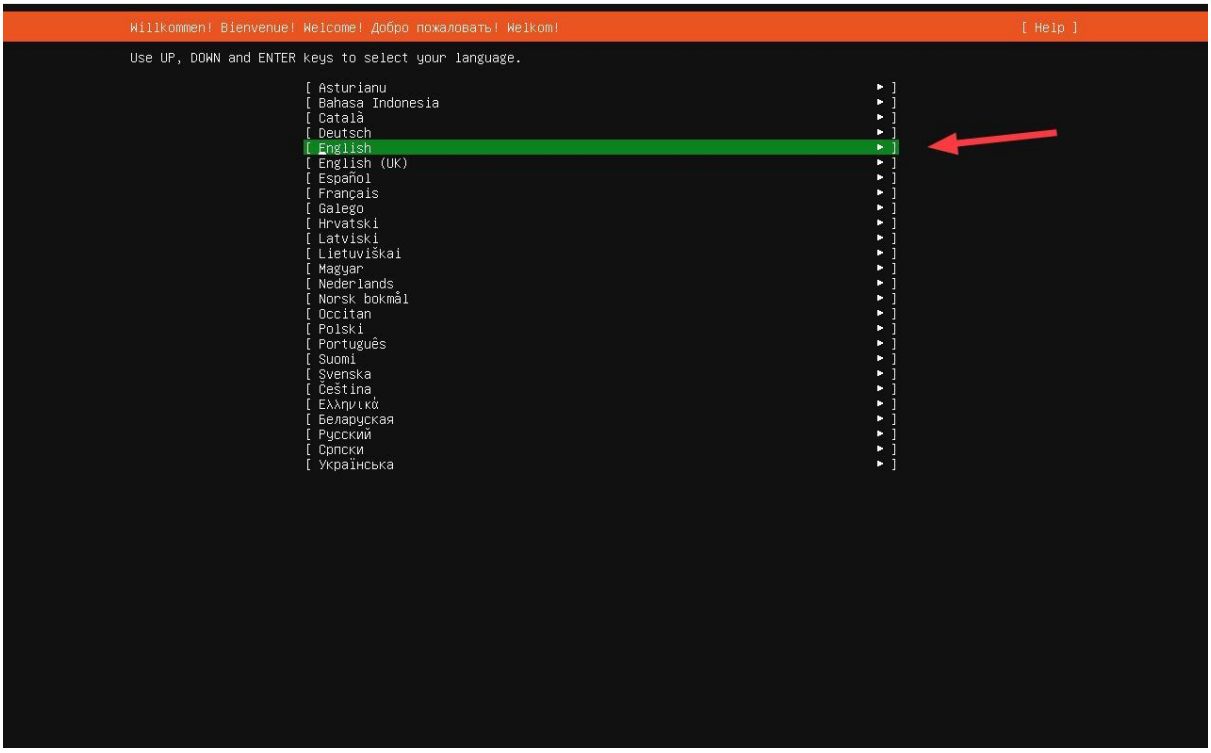

Oracle VM VirtualBox Manager
File
Machine
Help
Tools
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UbuntuServer-LTS
Running
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2.6
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kali-linux-2025.1a-virtualbox-amd64
Running


virtualbox.org/wiki/Downloads?utm_source=chatgpt.com
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Create Virtual Machine
Hardware
You can modify virtual machine's hardware by changing amount of RAM and virtual CPU count.
Enabling EFI is also possible.
Base Memory:
2048 MB
4 MB
16384 MB
Processors:
1
1 CPU
16 CPUs
Enable EFI (special OSes only)
Help
Back
Next
Cancel


Oracle VM VirtualBox Manager
File
Machine
Help
Tools
Ubuntu 22.04
Powered Off
New
Settings
Discard
Start
Start
General
Name: Ubuntu 22.04
Operating System: Ubuntu (64-bit)
System
Base Memory: 4096 MB
Boot Order: Floppy, Optical, Hard Disk
Acceleration: VT-x/AMD-V, Nested Paging, KVM
Paravirtualization
Preview
Ubuntu 22.04
Display
Video Memory: 30 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Secondary Device 0: [Optical Drive] ubuntu-22.04.3-desktop-amd64.iso (4.69 GB)
Controller: SATA
SATA Port 0: Ubuntu 22.04.vdi (Normal, 18.00 GB)


Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to:
Internal Network
Name:
pmat-internalnet
Advanced
Adapter Type:
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode:
Allow VMs
MAC Address:
0800276C1477
Cable Connected
OK
Cancel
Help


Willkommen! Bienvenue! Welcome! Добро пожаловать! Welkom!
[Help]
Use UP, DOWN and ENTER keys to select your language.
[Asturianu ▸]
[Bahasa Indonesia ▸]
[Català ▸]
[Deutsch ▸]
[English ▸]
[English (UK) ▸]
[Español ▸]
[Français ▸]
[Galego ▸]
[Hrvatski ▸]
[Latviski ▸]
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